

The Sturmian–Mahler Edge of the Accelerated Collatz Realizer Problem

Width-One Words, a 2-adic Hecke–Mahler Point, and an Effective Approximation Target

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June 2026

Abstract

We identify the rigid, minimally aperiodic edge of the confined-word realizer problem for the accelerated $3x + 1$ map with a single 2-adic object, and we locate that object precisely within the current Hecke–Mahler transcendence literature. The valuation words of width-one drift are exactly the mechanical (Sturmian) words of slope $\alpha = \log_2 3$, and under the Terras–Everett correspondence each such infinite word determines a 2-adic integer

$$\Xi_{\alpha,\beta} = -\frac{1}{3} \sum_{i \geq 0} 3^{-i} 2^{\lfloor i\alpha + \beta \rfloor} \in \mathbb{Z}_2,$$

the value at $(1/3, 2)$, taken 2-adically, of a Hecke–Mahler series. Least realizers of Sturmian prefixes are exactly the least positive residues of $\Xi_{\alpha,\beta}$ modulo 2^{S_N+1} . We prove: (i) an unconditional linear realizer floor on the edge, and that $\Xi_{\alpha,\beta}$ is never a rational integer (no integer orbit has an exactly mechanical valuation word); (ii) that any effective 2-adic irrationality measure for $\Xi_{\alpha,\beta}$ with exponent μ yields an exponential realizer floor $r(D_N) \gg 2^{S_N/\mu}$; (iii) a 2-adic domain lemma showing that the continued-fraction functional-equation orbit of the evaluation point of Bugeaud–Laurent stays strictly inside the 2-adic convergence domain at every stage, with unit-denominator corrections; (iv) a resonance dichotomy: the Collatz evaluation point lies on the *boundary* of the archimedean domain of every relevant transcendence theorem and in the *interior* of the 2-adic domain, so the failure of the archimedean theory to reach $\Xi_{\alpha,\beta}$ is relocation, not degeneracy; and (v) an exact equivalence between effective 2-adic measures and sublinear zero-run laws in the Hensel expansion of $\Xi_{\alpha,\beta}$. We position $\Xi_{\alpha,\beta}$ against the theorems of Mahler, Loxton–van der Poorten, Bugeaud–Laurent, Luca–Ouaknine–Worrell, and Masser, stating in each case the hypothesis that excludes our point, and consolidate the route into a single open problem: a 2-adic Mahler method along the Hecke–Mahler chain, whose place-free components are available and whose research core is a 2-adic zero estimate. No proof of irrationality of $\Xi_{\alpha,\beta}$, of an effective measure, of the Linear Confined-Realizer Floor (LCF), of the Effective Occupation Conjecture (EOC), or of the Collatz conjecture is claimed.

1 Introduction and non-claims

The accelerated Collatz map on odd integers is

$$T(m) = \frac{3m+1}{2^{v_2(3m+1)}}, \quad v_2 = 2\text{-adic valuation.}$$

An orbit emits a *valuation word* $D = (d_0, \dots, d_{N-1})$, $d_i = v_2(3m_i + 1) \geq 1$, with discharge $S_j = \sum_{i < j} d_i$ and drift $R_j = S_j - j\alpha$, $\alpha = \log_2 3$. By the Terras–Everett correspondence [1, 2, 3] each finite word has a least positive odd realizer $r(D)$. The companion notes [20, 19] formulate the Linear Confined-Realizer Floor (LCF) — a Collatz-sufficient linear lower bound on $r(D)$ for drift-confined words — prove sector floors, and localize the remaining obstruction to deep, aperiodic, sub-line words.

This note studies the opposite, *rigid* extreme of the confined landscape: words whose drift is confined to an interval of width less than one. These are exactly the mechanical (Sturmian) words of slope α , the minimal aperiodicity class immediately beyond the periodic words already priced by block-repetition costs in [20]. We show that this edge is governed by one concrete object of 2-adic Diophantine approximation, a Hecke–Mahler value $\Xi_{\alpha,\beta} \in \mathbb{Z}_2$ whose small positive residues are precisely the least realizers of Sturmian prefixes; we prove the unconditional facts available about it; and we situate it exactly relative to the active transcendence literature, whose recent advances [9, 10, 11] reached, at the archimedean place, the strength this edge would require at the 2-adic place.

Non-claim. This note does not prove the Collatz conjecture, does not prove LCF, and does not prove any exponential occupation floor (EOC). It does not address the sub-line, high-complexity residual core identified in [20], where the empirical realizer minimizers live. It does not prove irrationality of $\Xi_{\alpha,\beta}$ or any effective 2-adic measure, and it does not assert that any existing theorem applies to $\Xi_{\alpha,\beta}$ — the audit of Section 9 shows precisely that none does, for stated hypothesis-level reasons. The contribution is a reduction (an effective 2-adic measure for $\Xi_{\alpha,\beta}$ implies an exponential floor on the edge), the unconditional facts of Sections 5–7, and the positioning of the edge within the current literature.

2 Finite words and Terras–Everett realizers

For a word D of length N define the carry

$$C_N(D) = \sum_{i < N} 3^{N-1-i} 2^{S_i}, \quad C_0 = 0, \quad C_{j+1} = 3C_j + 2^{S_j},$$

so that if the word of m_0 begins with D then, by induction on j ,

$$m_j = \frac{3^j m_0 + C_j(D)}{2^{S_j}} \quad (0 \leq j \leq N). \quad (1)$$

Lemma 2.1 (Terras–Everett class). *For every word D of length N , the set of odd $m \geq 1$ whose word begins with D is the single nonempty residue class*

$$\{m : 3^N m + C_N(D) \equiv 2^{S_N} \pmod{2^{S_N+1}}\},$$

equivalently $m \equiv 3^{-N}(2^{S_N} - C_N(D)) \pmod{2^{S_N+1}}$.

Proof. By induction on N (as in [20]): the condition $d_N = v$ refines the level- N class by the single congruence $3m_N + 1 \equiv 2^v \pmod{2^{v+1}}$, affine in m_0 with odd leading coefficient. If m has word prefix D then m_N in (1) is an odd integer, which is exactly the displayed congruence; thus the word class is contained in the displayed class, and both are single classes modulo 2^{S_N+1} , hence equal. $\square$

Definition 2.2. $r(D)$ is the least positive element of the class of Lemma 2.1. It is non-decreasing under prefix extension (classes refine).

3 Width-one words are mechanical

Proposition 3.1 (Width one = mechanical). *Let $(S_j)_{j \geq 0}$ be integers with $S_0 = 0$ and $\sup_j (S_j - j\alpha) - \inf_j (S_j - j\alpha) < 1$. Then there is β with $S_j = \lfloor j\alpha + \beta \rfloor$ for all j , hence $d_j = S_{j+1} - S_j \in \{1, 2\}$.*

Proof. Set $\beta = \sup_j (S_j - j\alpha)$. For every j , $S_j \in (j\alpha + \beta - 1, j\alpha + \beta]$, a half-open interval of length 1 containing exactly one integer, namely $\lfloor j\alpha + \beta \rfloor$ (if $j\alpha + \beta \in \mathbb{Z}$ for some j , which can happen for at most one j by irrationality of α , the supremum is attained there and the value still equals the floor). The letter range follows from $1 < \alpha < 2$. $\square$

Remark 3.2 (The Sturmian identifications and two factor exclusions). The sequence $b_j = d_j - 1 \in \{0, 1\}$ is the mechanical word of slope $\alpha - 1 = \log_2(3/2)$ and intercept β , i.e. a Sturmian word [4, 5]. Width-one drift words, mechanical valuation words, and Sturmian codings are the same rigid edge in three coordinates: valuation slope α , letter frequencies $(2 - \alpha, \alpha - 1)$, and parity ones-density $1/\alpha$. The Sturmian slope $\alpha - 1 \approx 0.585$ exceeds $1/2$, so the minority letter $d = 1$ is isolated and the maximal run of $d = 2$ has length $\lfloor (\alpha - 1)/(2 - \alpha) \rfloor = 2$; hence *the factor (1, 1) never occurs and the factor (2, 2, 2) never occurs*, for any intercept β . Throughout, finite statements refer to prefixes D_N of a fixed infinite mechanical word with parameters (α, β) ; “Sturmian prefix” means such a prefix.

4 The 2-adic point $\Xi_{\alpha, \beta}$

Fix $\beta \in [0, 1)$, let $S_i = \lfloor i\alpha + \beta \rfloor$ (so $S_0 = 0$), let D_∞ be the corresponding infinite mechanical word, and let a_N be the residue of Lemma 2.1 for the prefix D_N .

Theorem 4.1 (Construction and identification). *The series*

$$\Xi_{\alpha, \beta} = -\frac{1}{3} \sum_{i \geq 0} 3^{-i} 2^{S_i} = -\sum_{i \geq 0} 3^{-i-1} 2^{\lfloor i\alpha + \beta \rfloor}$$

converges in $\mathbb{Z}_2$, and $\Xi_{\alpha, \beta} \equiv a_N \pmod{2^{S_N+1}}$ for every $N \geq 1$. Equivalently $\Xi_{\alpha, \beta}$ is the unique element of the nested intersection of the Terras–Everett classes of the prefixes D_N — the 2-adic realizer of the infinite word. It is the value of the Hecke–Mahler series $f_{\alpha, \beta}(x, y) = \sum_{i \geq 0} x^i y^{\lfloor i\alpha + \beta \rfloor}$ at $(x, y) = (1/3, 2)$, multiplied by $-1/3$, with convergence taken 2-adically.

Proof. The i -th term has 2-adic valuation $S_i \rightarrow \infty$ while 3^{-i-1} is a 2-adic unit, so the series converges in $\mathbb{Z}_2$. By Lemma 2.1, $a_N \equiv 3^{-N} 2^{S_N} - \sum_{i < N} 3^{-i-1} 2^{S_i} \pmod{2^{S_N+1}}$, using $3^{-N} C_N = \sum_{i < N} 3^{-i-1} 2^{S_i}$. Hence

$$\Xi_{\alpha, \beta} - a_N \equiv -3^{-N} 2^{S_N} - \sum_{i \geq N} 3^{-i-1} 2^{S_i} = -3^{-N-1} 2^{S_N+2} - \sum_{i \geq N+1} 3^{-i-1} 2^{S_i} \pmod{2^{S_N+1}},$$

combining the index- N terms via $3^{-N}2^{S_N} + 3^{-N-1}2^{S_N} = 3^{-N-1}2^{S_N}(3+1) = 3^{-N-1}2^{S_N+2}$. Every remaining term has valuation $\geq S_N+2$ or $\geq S_{N+1} \geq S_N+1$, so $\Xi_{\alpha,\beta} \equiv a_N \pmod{2^{S_N+1}}$. Uniqueness of the nested-class limit is the divergence of the moduli. $\square$

Remark 4.2 (Periodic sanity checks). The same series on periodic discharge sequences reproduces the block fixed points of [20]: $S_i = i$ (all $d_i = 1$) gives $-\frac{1}{3} \sum (2/3)^i = -1$; $S_i = 2i$ (all $d_i = 2$) gives $-\frac{1}{3} \sum (4/3)^i = +1$; the word $(1, 2)^\infty$ gives -5 (geometric sums evaluated 2-adically, e.g. $\sum_i 2^i 3^{-i-1} = 1$ in $\mathbb{Z}_2$). The Sturmian point is the genuinely new, minimally aperiodic member of a family whose periodic members are the rational cycle points.

5 Least realizers, the reduction, and the linear floor

Proposition 5.1 (Least realizers are small residues). *For every $N \geq 1$, $r(D_N) = \min\{y \geq 1 : y \equiv \Xi_{\alpha,\beta} \pmod{2^{S_N+1}}\}$. Hence, for $\varepsilon \in (0, 1]$, $r(D_N) < \varepsilon 2^{S_N}$ holds iff there is an integer y with $0 < y < \varepsilon 2^{S_N}$ and $|\Xi_{\alpha,\beta} - y|_2 \leq 2^{-S_N-1}$, i.e. a positive integer with $|y|_\infty \cdot |\Xi_{\alpha,\beta} - y|_2 < \varepsilon/2$.*

Proof. Immediate from Theorem 4.1 and the definition of r . $\square$

Small least realizers on the Sturmian edge are therefore exactly small positive residues of a single 2-adic Hecke–Mahler value along the modulus sequence 2^{S_N+1} : a simultaneous archimedean–2-adic approximation problem.

Theorem 5.2 (Effective measure implies exponential edge floor). *Suppose there exist $C > 0$ and $\mu < \infty$ with $|\Xi_{\alpha,\beta} - y|_2 \geq C y^{-\mu}$ for every integer $y \geq 1$. Then $r(D_N) \geq (2C)^{1/\mu} 2^{S_N/\mu}$ for all $N \geq 1$.*

Proof. Apply the hypothesis to $y = r(D_N)$, which satisfies $|\Xi_{\alpha,\beta} - y|_2 \leq 2^{-S_N-1}$ by Proposition 5.1: $C r^{-\mu} \leq 2^{-S_N-1}$, i.e. $r^\mu \geq 2C \cdot 2^{S_N}$. $\square$

Remark 5.3 (Strength levels). Any finite effective μ already yields an exponential floor $2^{S_N/\mu}$, far beyond the linear LCF scale on this edge. The strong prediction $\log_2 r(D_N) \geq S_N - O(\log N)$ of Section 10 would correspond to a near-optimal $\mu = 1 + o(1)$ and is *not* claimed; the realistic open target is any finite μ (Open Problem 2).

Lemma 5.4 (Edge injectivity). *Let m realize the Sturmian prefix D_N and let (i, k) , $0 \leq i < k \leq N$, be its first state repetition. Then m_i lies on a positive cycle and the word is $(k-i)$ -periodic from index i ; if that cycle is the trivial cycle $\{1\}$ then $i \geq N-2$. Hence either $m_0, \dots, m_{N-2}$ are pairwise distinct, or the orbit meets a nontrivial positive cycle.*

Proof. A first repetition puts the orbit on a cycle from index i with periodic word thereafter. The trivial cycle has word $(2)^\infty$; if $i \leq N-3$ the prefix would contain the factor $(2, 2, 2)$, excluded for mechanical slope- α words (Remark 3.2). Any other positive cycle is nontrivial. $\square$

Theorem 5.5 (Unconditional linear floor on the edge). *Let D_N be a Sturmian prefix with intercept $\beta \in [0, 1)$, $N \geq 5$. Then either the orbit of $r(D_N)$ meets a nontrivial positive Collatz cycle, or*

$$r(D_N) \geq \frac{3}{4} 2^{\beta-6/5} (N-3) \geq 0.32 (N-3).$$

Proof. The drift satisfies $R_j = \lfloor j\alpha + \beta \rfloor - j\alpha \in (\beta - 1, \beta]$, a band with floor depth $w = 1 - \beta$ and ceiling $c = \beta$. By Lemma 5.4, in the first alternative the realizer is injective for D_{N-2} , and the Band-Confined Linear Floor of [20] gives $r(D_N) \geq r(D_{N-2}) \geq \frac{3}{4} 2^{-(6w+c)/5} (N-3)$ with $(6w+c)/5 = 6/5 - \beta$. $\square$

Corollary 5.6 (No integer realizes a mechanical word). $\Xi_{\alpha,\beta} \notin \mathbb{Z}$: no integer orbit of T has an exactly mechanical valuation word of slope α . This is unconditional.

Proof. Suppose $\Xi_{\alpha,\beta} = m \in \mathbb{Z}$. If $m \geq 1$, then m realizes every prefix D_N . If the orbit is injective on all initial segments, Theorem 5.5 forces $m \geq 0.32(N-3) \rightarrow \infty$, absurd; otherwise the orbit meets a cycle, so the *infinite* word of m is eventually periodic, contradicting Sturmian aperiodicity. If $m \leq -1$: the exact occupation-state identity $m_j = 2^{-R_j} (m + \frac{1}{3} \sum_{i < j} 2^{R_i})$ [20], with $R_i \in (\beta - 1, \beta]$ bounded, forces $\sum_{i < j} 2^{R_i} \geq j 2^{\beta-1} \rightarrow \infty$, so $m_j > 0$ for large j , absurd. Finally $m = 0$ is not odd. $\square$

Remark 5.7. For high-placed bands (β near 1) the constant $\frac{3}{4} 2^{\beta-6/5}$ reaches $0.65 > 1/(3 \ln 2) \approx 0.481$, the LCF constant: on that slice of the edge the LCF inequality itself is proved for injective realizers.

6 The Hecke–Mahler chain and the resonance dichotomy

Definition 6.1. For slope $\gamma > 0$ and intercept ρ , set $f_{\gamma,\rho}(z_1, z_2) = \sum_{i \geq 0} z_1^i z_2^{\lfloor i\gamma + \rho \rfloor}$, so $\Xi_{\alpha,\beta} = -\frac{1}{3} f_{\alpha,\beta}(1/3, 2)$. Bugeaud–Laurent [9] use the normalization $F_{\theta,\rho}(z_1, z_2) = \sum_{k_1 \geq 1} \sum_{k_2=1}^{\lfloor k_1\theta + \rho \rfloor} z_1^{k_1} z_2^{k_2}$, $0 \leq \theta, \rho < 1$, θ irrational, and prove that $F_{\theta,\rho}$ takes transcendental values at every algebraic (z_1, z_2) with $0 < |z_1|, |z_1 z_2^\theta| < 1$, for all irrational slopes, together with an explicit continued-fraction expansion and irrationality exponent for $F_{\theta,\rho}(1/b, 1/a)$ at rational points strictly inside that domain.

Lemma 6.2 (Bridge between normalizations). For $0 \leq \rho < 1$, as an identity of formal series and of 2-adic values where both converge,

$$f_{\theta,\rho}(z_1, z_2) = \frac{1}{1 - z_1} + \frac{z_2 - 1}{z_2} F_{\theta,\rho}(z_1, z_2).$$

Proof. $\sum_{k_2=1}^M z_2^{k_2} = z_2(z_2^M - 1)/(z_2 - 1)$; sum over $k_1 \geq 1$ with $M = \lfloor k_1\theta + \rho \rfloor$ to get $F_{\theta,\rho} = \frac{z_2}{z_2 - 1} [(f_{\theta,\rho} - 1) - \frac{z_1}{1 - z_1}]$ (the $k_1 = 0$ term of f is $z_2^{\lfloor \rho \rfloor} = 1$), and solve. $\square$

The functional equations are continued-fraction-driven. With $\theta = [0; a_1, a_2, \dots]$ and Gauss iterates $\theta_{k+1} = \{1/\theta_k\}$, [9] display the homogeneous chain

$$F_{\theta_k,0}(z_1, z_2) = -F_{\theta_{k+1},0}(z_1^{a_{k+1}+1} z_2, z_1) + \frac{z_1^{a_{k+1}+1} z_2}{(1 - z_1^{a_{k+1}+1} z_2)(1 - z_1)}, \quad (2)$$

the inhomogeneous intercept being transported by the Ostrowski representation [9, §§2,4]. Our slope satisfies $1 < \alpha < 2$; the exact correction-free reduction $f_{\alpha,\beta}(x, y) = f_{\alpha-1,\beta}(xy, y)$ (from $\lfloor i\alpha + \beta \rfloor = i + \lfloor i(\alpha - 1) + \beta \rfloor$) brings it to slope $\theta = \alpha - 1 \in (0, 1)$, after which (2) applies.

Remark 6.3 (Not a d -Mahler value). $\Xi_{\alpha,\beta}$ satisfies no equation of the standard d -Mahler form $\sum_i a_i(x)f(x^{d^i}) = 0$; its functional-equation monoid is the continued-fraction monoid, not a fixed power map. The effective d -Mahler and automatic-number theories [15] therefore do not transfer, and the object lives on the Hecke–Mahler [7, 8, 13, 9] side of the theory.

Definition 6.4 (2-adic convergence domain). For slope $\gamma > 0$, $\mathcal{D}_\gamma^{(2)} = \{(z_1, z_2) \in \mathbb{Q}_2^2 : v_2(z_1) + \gamma v_2(z_2) > 0\}$; on this domain $f_{\gamma,\rho}$ and $F_{\gamma,\rho}$ converge in $\mathbb{Q}_2$, the term valuations growing linearly.

Theorem 6.5 (Domain persistence; resonance dichotomy). *Start from $(x, y) = (1/3, 2)$ and apply the initial correction-free shift to $(z_1, z_2) = (2/3, 2)$, valuation vector $(u, v) = (1, 1)$. The chain move of (2) acts on valuation vectors by $(u, v) \mapsto (a_{k+1}u + v, u)$. Then at every stage $k \geq 0$:*

1. (2-adic interior) *both entries are ≥ 1 , so $v_2(z_1) + \theta_k v_2(z_2) \geq 1 > 0$ and the evaluation point lies strictly inside $\mathcal{D}_{\theta_k}^{(2)}$; moreover every correction denominator in (2) is a 2-adic unit ($v_2(z_1^{a_{k+1}} z_2) = a_{k+1}u + v \geq 2$ and $v_2(z_1) = u \geq 1$);*
2. (archimedean boundary) $|z_1 z_2^{\theta_k}|_\infty = 1$, *so the point lies on the boundary of the archimedean domain of definition at every stage.*

Consequently every series arising from $f_{\alpha,\beta}(1/3, 2)$ along the continued-fraction algorithm of α converges 2-adically with unit-denominator corrections, while no complex value exists at any stage.

Proof. (i) With $u, v \geq 1$ the move gives entries $a_{k+1}u + v \geq 2$ and $u \geq 1$, so “both entries ≥ 1 ” is preserved by induction from $(1, 1)$; then $v_2(z_1) + \theta_k v_2(z_2) \geq v_2(z_1) \geq 1$, and a 2-adic integer of positive valuation t makes $1 - t$ a unit. (ii) Initially $|z_1 z_2^{\theta_0}| = |2/3| \cdot |2|^{\alpha-1} = \frac{2}{3} \cdot 2^{\alpha-1} = \frac{2}{3} \cdot \frac{3}{2} = 1$. If it holds at stage k , then with $z'_1 = z_1^{a_{k+1}} z_2$, $z'_2 = z_1$, $\theta' = 1/\theta_k - a$ ($a = a_{k+1}$): $|z'_1 (z'_2)^{\theta'}| = |z_1|^{a+\theta'} |z_2| = |z_1|^{1/\theta_k} |z_2| = |z_2|^{-1} |z_2| = 1$, using the resonance $|z_1| = |z_2|^{-\theta_k}$. The convergence claim is the domain definition. $\square$

Remark 6.6 (The failure mode is relocation, not degeneracy). At the resonant point the terms have archimedean size $\Theta(1)$ (Theorem 4.1): the series has *no* complex value, at any stage. The archimedean theorems of [7, 8, 9] therefore do not fail to handle $\Xi_{\alpha,\beta}$ through a vanishing subtlety; their hypotheses exclude it because their object does not exist there. By Theorem 6.5(i) the same point is interior to the 2-adic domain at every stage, so a 2-adic Mahler method, if constructed, faces no domain obstruction here. The resonance $xy^\alpha = 1$ — the identity that makes $\Xi_{\alpha,\beta}$ encode Collatz drift — is the boundary of the wrong place and the interior of the right one.

7 Digit truncation and the measure/zero-run equivalence

Write $\Xi_{\alpha,\beta} = \sum_{j \geq 0} \varepsilon_j 2^j$, $\varepsilon_j \in \{0, 1\}$, for the Hensel expansion.

Proposition 7.1 (Realizers are Hensel truncations). *$r(D_N) = \sum_{j=0}^{S_N} \varepsilon_j 2^j$, the integer formed by the first $S_N + 1$ Hensel bits of $\Xi_{\alpha,\beta}$; it is odd ($\varepsilon_0 = 1$). If $2^k \leq r(D_N) < 2^{k+1}$ then $\varepsilon_{k+1} = \dots = \varepsilon_{S_N} = 0$, a zero-run of length $S_N - k$ below depth $S_N + 1$, with $S_N - k = \lceil E_N \rceil + O(1)$, $E_N := S_N - \log_2 r(D_N)$.*

Proof. The truncation is a positive odd integer $\equiv \Xi_{\alpha,\beta} \pmod{2^{S_N+1}}$ and $< 2^{S_N+1}$, hence the least positive residue $r(D_N)$ by Proposition 5.1. Its bit-length is $k+1$, so the higher bits up to S_N vanish, and $E_N \in (S_N - k - 1, S_N - k]$. $\square$

Proposition 7.2 (Measure $\Leftrightarrow$ sublinear zero-run law). *Fix $\mu \in (1, \infty)$. The following are equivalent up to the constants: (i) $|\Xi_{\alpha,\beta} - y|_2 \geq C y^{-\mu}$ for every integer $y \geq 1$, some $C > 0$; (ii) every zero-run in the Hensel expansion of $\Xi_{\alpha,\beta}$ ending at depth k has length at most $(1 - \frac{1}{\mu})k + C'$, some $C' \geq 0$.*

Proof. (i) $\Rightarrow$ (ii): a zero-run of length ℓ ending at depth k gives a truncation $y_k < 2^{k-\ell}$ with $|\Xi_{\alpha,\beta} - y_k|_2 \leq 2^{-k}$, so $C 2^{-\mu(k-\ell)} \leq C y_k^{-\mu} \leq 2^{-k}$ and $\ell \leq (1 - \frac{1}{\mu})k + \frac{1}{\mu} \log_2 C^{-1}$. (ii) $\Rightarrow$ (i): for an integer $y \geq 1$ with agreement depth k ($|\Xi_{\alpha,\beta} - y|_2 = 2^{-k}$), one has $y \geq y_k$ and, by (ii), y_k has bit-length $\geq k/\mu - C'$, so $y \geq 2^{k/\mu - C' - 1}$, whence $k \leq \mu(\log_2 y + C' + 1)$ and $|\Xi_{\alpha,\beta} - y|_2 \geq 2^{-\mu(C'+1)} y^{-\mu}$. $\square$

Remark 7.3 (Calibration). A *finite* exponent μ is a sublinear zero-run law (runs $\leq (1 - \frac{1}{\mu}) \cdot \text{depth}$) — a digit-theoretic statement, strictly weaker than the $O(\log)$ law, which is $\mu = 1 + o(1)$ and normality-adjacent. Any finite μ already yields the exponential edge floor of Theorem 5.2. A 2-adic Mahler method, by analogy with the archimedean transcendence-measure output, would realistically deliver a finite μ , not the sharp law.

8 Convergent-block windows

Let p_k/q_k be the convergents of $\alpha = \log_2 3$, with denominators $1, 2, 5, 12, 41, 53, 306, \dots$. Sturmian words are concatenations of standard blocks governed by this continued fraction and contain powers of the standard blocks with exponents governed by the partial quotients [5, 6]. The standard block B_k has length q_k and discharge p_k , with $|2^{p_k} - 3^{q_k}| = 3^{q_k} |2^{p_k - q_k \alpha} - 1| \asymp 3^{q_k} / q_{k+1}$, small in relative terms precisely because p_k/q_k is a convergent; its block-cost loss satisfies $\lambda_k \leq \log_2 |2^{p_k} - 3^{q_k}| \approx p_k - \log_2 q_{k+1}$.

Proposition 8.1 (Windowed exponential floors at convergent scales). *If the Sturmian prefix D_N contains $s \geq 2$ consecutive copies of a convergent block B_k (or a rotation) beginning at index a , then by the block-cost theorems of [20], and using that edge entrance depth is below 1,*

$$r(D_N) \geq 2^{-1} 2^{s p_k - \lambda_k} - \frac{2}{3} N$$

(unless the entrance state lies on a cycle of the matching orientation), with $\lambda_k \leq \log_2 |2^{p_k} - 3^{q_k}|$.

The qualitative reading: where α has a large partial quotient, the edge realizers are forced exponentially large over the corresponding window by block-cost alone; where the partial quotients are small, periodic pricing is weak and the approximation route of Section 5 is the only visible mechanism. The two are complementary in the way continued-fraction arithmetic suggests. No global floor follows from these windows alone, since ranges between convergent scales are not covered.

9 Positioning: the literature gate, by hypothesis

The Hecke–Mahler transcendence literature is active and recent. For each relevant theorem we state the hypothesis that excludes $\Xi_{\alpha,\beta}$, not an impression.

- **Mahler 1929 [7]; Loxton–van der Poorten 1977 [8].** Transcendence of homogeneous ($\rho = 0$) Hecke–Mahler values at algebraic points of the open domain $|z_1| < 1$, $|z_1 z_2^\theta| < 1$; complex place. Excluded by Theorem 6.5(ii): our point is on the boundary, where no complex value exists.
- **Bugeaud–Laurent 2023 [9].** Transcendence of the inhomogeneous $F_{\theta,\rho}$ at all algebraic points with $0 < |z_1|, |z_1 z_2^\theta| < 1$ strictly, for all irrational slopes, plus explicit continued-fraction expansions and irrationality exponents for $F_{\theta,\rho}(1/b, 1/a)$ at rational points inside the domain. The closest theorem in existence: effective Hecke–Mahler approximation is a theorem at the archimedean place. Excluded for us by the same strict domain hypothesis (our point is on the boundary, and is $(1/3, 2)$ rather than a $(1/b, 1/a)$ interior point); the effectivity mechanism — the continued fraction of the real value — has no 2-adic counterpart, where integer approximation is governed by Hensel truncations (Proposition 7.2).
- **Luca–Ouaknine–Worrell 2022/2025 [11, 10].** Transcendence of $\sum_n f(\lfloor n\theta + \rho \rfloor) \beta^{-n}$ for $f \in \mathbb{Z}[x]$ nonconstant and algebraic $|\beta| > 1$, the 2022 argument via the p -adic Subspace Theorem [14]. The values are archimedean with coefficients of polynomial growth; our coefficients $2^{\lfloor i\alpha + \beta \rfloor}$ grow like 3^i , so the terms are $\Theta(1)$ archimedeanly — the resonance again — and the “ p -adic” in the method concerns auxiliary places, not the place of the value. Not applicable.
- **Masser 1982 [12]; Masser 1999 [13].** The vanishing theorem for power series is the standard non-vanishing input for single-transformation several-variable Mahler systems; the 1999 paper gives algebraic independence of Hecke–Mahler values for *quadratic* slopes at the complex place. Our slope $\log_2 3$ is not quadratic and our place is not archimedean. The vanishing component, however, is substantially arithmetic/combinatorial, and its flexible subspace-theorem form (used in Mahler’s method by [15]) is the component of the archimedean theory most plausibly transferable to a 2-adic method.

Remark 9.1 (On partial quotients). The classical easy/hard split is, for approximation-based transcendence, *reversed* from the Mahler-method split: unbounded partial quotients is the easy case for approximation, bounded the case requiring the machinery. Since [9] covers all irrational slopes, no standing hypothesis on the (unknown) partial quotients of $\log_2 3$ is needed, and a 2-adic transfer should be planned for all slopes.

10 Computations

Exact integer arithmetic throughout: mechanical floors $\lfloor i\alpha + \beta \rfloor$ by comparison of powers of 2 and 3; realizers as Hensel truncations of $\Xi_{\alpha,\beta}$ computed modulo a large power of 2, cross-checked against exact Terras–Everett lifting. Finite-range diagnostics; nothing below is a proof.

- *Sturmian floor confirmed.* The factor complexity of the mechanical valuation words equals $p(n) = n + 1$ over the tested ranges ($n \leq 10$, prefixes of length 400).
- *Residuals.* For $\beta \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ and $N \leq 2500$, the residuals $E_N = S_N - \log_2 r(D_N)$ oscillate near 0 at generic N , with isolated spikes; the maximum over $N \leq 2500$ is $E \approx 14.4$ for $\beta = 0$ (near $N = 723 \approx 2.4 \times 306$, not increasing between $N = 1000$ and $N = 2500$) and $E \approx 11.8$ for $\beta = \frac{1}{2}$.

- *Spikes sit at convergent scales.* Large residuals cluster near multiples of the convergent denominators 12, 41, 53, 306; peak heights are consistent with $\approx \log_2 q_{k+1} + O(1)$ from the convergent-block losses of Section 8, and equally consistent with the random-digit background (longest zero-run among M Hensel digits $\sim \log_2 M \approx 12$ at this depth). The data does not, and beyond constants cannot, distinguish the deterministic continued-fraction mechanism from chance.
- *No power-law drift.* No $\sqrt{N}$ or linear residual growth is observed; residuals return near 0 between spikes.
- *Methodological caution.* An earlier peak-location diagnostic that matched peaks to multiples of *all* convergent denominators was vacuous (small denominators make every integer a near-multiple); the reliable diagnostics are residual size, zero-run length, and superlogarithmic anomaly growth, none observed.

These observations support the approximation framing and falsify nothing; they prove nothing. A residual drifting like a power of N , or a spike at a non-convergent scale, in extended ranges would demote the route.

11 Open problems

Open Problem 1 (Irrationality). Prove $\Xi_{\alpha,\beta} \notin \mathbb{Q}$.

Remark 11.1 (Not automatic). Irrationality does *not* follow from Sturmian aperiodicity: “rational 2-adic $\Rightarrow$ eventually periodic valuation word” is the open periodicity conjecture [3], not a theorem. If $\Xi_{\alpha,\beta} = p/q$ in lowest terms with q odd, then $n_j = qT^j(\Xi_{\alpha,\beta})$ is an integer orbit of $n \mapsto (3n + q)/2^{v_2(3n+q)}$ realizing the mechanical word: rationality of $\Xi_{\alpha,\beta}$ is equivalent to a mechanical integer orbit in a generalized $3x + q$ system. Corollary 5.6 settles only $q = 1$.

Open Problem 2 (Effective measure via a 2-adic Mahler method). Construct a Mahler-method irrationality (or transcendence) argument for values of the chain (2) at points of the 2-adic domain $\mathcal{D}^{(2)}$, applicable to $\Xi_{\alpha,\beta} = -\frac{1}{3}f_{\alpha,\beta}(1/3, 2)$, and extract an effective 2-adic measure $|\Xi_{\alpha,\beta} - y|_2 \geq Cy^{-\mu}$. The place-free or already-settled components are: the functional equations (2) with the bridge of Lemma 6.2; the domain and unit-denominator statements of Theorem 6.5; the auxiliary-polynomial construction (place-independent linear algebra); a Liouville-type lower bound (valid at every place); and the 2-adic contraction of the orbit (term valuations grow linearly along the chain). The research core is the *zero estimate*: a non-vanishing theorem for the auxiliary function along the chain’s monomial substitutions, in the 2-adic domain; the most plausible source is the arithmetic (subspace-theorem) form of the vanishing theory [15]. By Theorem 5.2 any finite μ yields $r(D_N) \gg 2^{S_N/\mu}$.

Remark 11.2 (Precedent for the place transfer). The genre is not speculative. The transcendence of p -adic values of the modular function (the p -adic analogue of Mahler’s archimedean conjecture) was posed by Manin and proved, together with the archimedean case, by Barré-Sirieix, Diaz, Gramain and Philibert [16]. A p -adic-place analogue of an archimedean Mahler-flavored theorem has been a named conjecture before and was eventually proved. This is cited for the *shape* of Open Problem 2, not as evidence of its truth or tractability.

Open Problem 3 (Strong residual / zero-run law). Determine whether $E_N = S_N - \log_2 r(D_N) = O(\log N)$ for all Sturmian prefixes, or any subpower bound $E_N = O(S_N^\theta)$, $\theta < 1$. By Proposition 7.2 the $O(\log)$ form is normality-adjacent and should be regarded as hard.

12 Conclusion

Non-claim. This note controls or reframes the rigid, width-one, minimally aperiodic edge only. The empirical minimizers of the confined realizer problem are sub-line, deep, and complexity-saturated [20]; nothing here touches them. No proof of the Collatz conjecture, of LCF, or of EOC is claimed.

Provable now, and proved here: the identification of the edge with the 2-adic Hecke–Mahler point $\Xi_{\alpha,\beta}$ (Theorem 4.1); the unconditional linear floor and integer exclusion $\Xi_{\alpha,\beta} \notin \mathbb{Z}$ (Theorem 5.5, Corollary 5.6); the reduction of the strong edge floor to an effective 2-adic measure (Theorem 5.2); the domain lemma and boundary/interior dichotomy (Theorem 6.5); and the measure/zero-run equivalence (Proposition 7.2). Open: irrationality of $\Xi_{\alpha,\beta}$, any effective measure, and the consolidated 2-adic Mahler method of Open Problem 2, into which the literature gate collapses. Out of reach as a theorem: the $O(\log N)$ residual law, which is normality-adjacent. The honest headline is that the edge problem is now exactly located in the active literature — one place-transfer away from a theory that reached, in 2023–2025, precisely the strength the edge would need at the archimedean place, with the Collatz evaluation point sitting on the boundary of that theory’s world and in the interior of the 2-adic one, where no such theory yet exists.

Acknowledgments

Adversarial auditing, the literature search, and computational verification were assisted by Claude (Anthropic). All mathematical claims, their framing, and any errors are the author’s.

Bibliographic note

References [7, 8, 13, 14, 9, 10, 11, 16] were verified against primary or publisher sources; [12, 15] are cited from standard secondary use. Companion-note DOIs other than [19] remain placeholders to be inserted at deposit.

References

- [1] R. Terras, A stopping time problem on the positive integers, *Acta Arithmetica* **30** (1976), 241–252.
- [2] C. J. Everett, Iteration of the number-theoretic function $f(2n) = n$, $f(2n + 1) = 3n + 2$, *Advances in Mathematics* **25** (1977), 42–45.
- [3] J. C. Lagarias, The $3x + 1$ problem and its generalizations, *American Mathematical Monthly* **92** (1985), 3–23.
- [4] M. Morse and G. A. Hedlund, Symbolic dynamics II. Sturmian trajectories, *American Journal of Mathematics* **62** (1940), 1–42.
- [5] M. Lothaire, *Algebraic Combinatorics on Words*, Chapter 2 (J. Berstel and P. Séébold, Sturmian words), Cambridge University Press, 2002.
- [6] F. Mignosi, Infinite words with linear subword complexity, *Theoretical Computer Science* **65** (1989), 221–242.

- [7] K. Mahler, Arithmetische Eigenschaften der Lösungen einer Klasse von Funktionalgleichungen, *Mathematische Annalen* **101** (1929), 342–366.
- [8] J. H. Loxton and A. J. van der Poorten, Arithmetic properties of certain functions in several variables III, *Bulletin of the Australian Mathematical Society* **16** (1977), 15–47.
- [9] Y. Bugeaud and M. Laurent, Transcendence and continued fraction expansion of values of Hecke–Mahler series, *Acta Arithmetica* **209** (2023), 59–90. doi:10.4064/aa220323-18-1; arXiv:2203.12901.
- [10] F. Luca, J. Ouaknine, and J. Worrell, Transcendence of Hecke–Mahler series, *Bulletin of the London Mathematical Society* **57** (2025), 1360–1368. doi:10.1112/blms.70033.
- [11] F. Luca, J. Ouaknine, and J. Worrell, On the transcendence of a series related to Sturmian words, *Annali della Scuola Normale Superiore di Pisa*, to appear; arXiv:2204.08268.
- [12] D. W. Masser, A vanishing theorem for power series, *Inventiones Mathematicae* **67** (1982), 275–296.
- [13] D. W. Masser, Algebraic independence properties of the Hecke–Mahler series, *Quarterly Journal of Mathematics (Oxford)* **50** (1999), 207–230.
- [14] H. P. Schlickewei, Die p -adische Verallgemeinerung des Satzes von Thue–Siegel–Roth–Schmidt, *Journal für die reine und angewandte Mathematik* **288** (1976), 86–105.
- [15] B. Adamczewski and C. Faverjon, Mahler’s method in several variables I: the theory of regular singular systems, preprint, arXiv:1809.04823.
- [16] K. Barré-Sirieix, G. Diaz, F. Gramain, and G. Philibert, Une preuve de la conjecture de Mahler–Manin, *Inventiones Mathematicae* **124** (1996), 1–9. doi:10.1007/s002220050044.
- [17] K. Nishioka, *Mahler Functions and Transcendence*, Lecture Notes in Mathematics 1631, Springer, 1996.
- [18] O. Rozier, The $3x + 1$ problem: a lower bound hypothesis, *Functiones et Approximatio Commentarii Mathematici* **56** (2017), 7–23.
- [19] E. De Jesús, A global occupation conjecture for the accelerated $3x + 1$ map, with an unconditional logarithmic-floor exclusion for injective orbits, technical note, Zenodo (2026). doi:10.5281/zenodo.20575276.
- [20] E. De Jesús, A word-level occupation–realizer reduction for the accelerated $3x + 1$ map, technical note, Zenodo (2026).<https://doi.org/10.5281/zenodo.20593925>